



THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK EXAMINATION

032/1

CHEMISTRY I

TIME: 3:00 HOURS

YEAR: 2026

INSTRUCTIONS

1. This paper consists of section **A, B** and **C** with a total of **eleven (11)** questions.
2. Answer **all** questions in section **A** and **B** and **two (02)** questions from section **C**
3. Section **A** carries **sixteen (16)** marks, section **B** **fifty-four (54)** marks and section **C** **thirty (30)** marks.
4. All writings must be in **blue** or **black** ink except drawing which must be **pencil**.
5. Cellular phones and any unauthorized materials are **not allowed** in the examination room.
6. Write your **Examination number** on every page of your answer booklet(s)
7. The following constants **may** be used:

Atomic masses: H= 1, C= 12, O= 16, S= 32

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SECTION A: (16 MARKS)
Answer all questions in this section

1. For each of the items (i)-(x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.
- (i) Which of these statements is true about isotopes of an element?
 A. The number of protons is the same, but the number of neutrons is different.
 B. The number of protons and neutrons is the same, but the number of electrons is different.
 C. The number of neutrons is the same, but the number of protons is different.
 D. The number of protons is the same, but electrons are added to the nucleus.
 E. The number of neutrons and electrons are the same.
- (ii) Which of the following transforms a mechanical energy to an electric energy?
 A. Hydroelectric power plant
 B. Solar panel
 C. An electric cooker
 D. An electric motor
 E. A motorcycle wheel
- (iii) Which oxides among the following are gaseous at room temperature?
 A. Copper oxide and iron oxide
 B. Carbon dioxide and copper oxide
 C. Carbon dioxide and Sulphur dioxide
 D. Sulphur dioxide and copper oxide
 E. Iron oxide and carbon dioxide
- (iv) Steam and methane can react and produce hydrogen according to the following equation. $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \longrightarrow \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$ The volume of methane needed to produce 150cm^3 of hydrogen at the same conditions will be;
 A. 50cm^3
 B. 25cm^3
 C. 100cm^3
 D. 75cm^3
 E. 150cm^3
- (v) Which of the following statements is not true about hydrogen gas?
 A. Is a reducing agent.
 B. Burns in air to form steam.
 C. Is a neutral gas, almost insoluble in water.
 D. Diffuses more rapidly than carbon dioxide.
 E. Is prepared by the action of dilute nitric acid on zinc metal.
- (vi) Which one of the following substances where its dissolution in water causes permanent hardness of water?
 A. Sodium carbonate
 B. Magnesium carbonate
 C. Calcium sulphate
 D. Sodium chloride
 E. Calcium carbonate
- (vii) What will happen as water is being added to an acid?
 A. The acid becomes more acidic and its pH goes down.
 B. The acid becomes neutral and its pH becomes 7.
 C. The acid becomes less acidic and its pH goes down.
 D. The acid becomes more acidic and its pH goes up.
 E. The acid becomes less acidic and its pH goes up.
- (viii) A Bunsen burner will produce a yellow flame when;
 A. The air hole is fully closed.
 B. The gas tap is fully opened.
 C. Sufficient gas is supplied to the Bunsen burner.
 D. The gas tap is partially opened.
 E. The air hole is fully opened.
- (ix) Which of the following sets of laboratory apparatus are used to measure volume of water?
 A. Thistle funnel, separation funnel and beaker
 B. Burette, pipette and measuring cylinder
 C. Crucible, Kipp's apparatus and volumetric flask
 D. Volumetric flask, distillation flask and test tube
 E. Test tube, beaker and gas jar
- (x) Form Four students in a certain school were doing scientific research about malaria disease in the Coast Region. Which among the following scientific procedures was used to accept or reject the hypothesis made?
 A. Data interpretation
 B. Experimentation
 C. Conclusion
 D. Hypothesis formulation
 E. Problem identification

2. Match the chemical composition of fire extinguisher in **List "A"** with the corresponding extinguisher in **List "B"** by writing the letter of the correct response beside the item number in the answer booklet provided.

S/N	LIST A:	LIST B:
(i)	The extinguisher composed of bromochlorodifluoromethane	A. Dry chemical extinguisher B. Fire blanket extinguisher C. APW extinguisher D. Carbon dioxide extinguisher E. Halon extinguisher F. ABC extinguisher G. Sand bucket extinguisher H. Wet chemical extinguisher I. Foam extinguisher
(ii)	The extinguisher composed of potassium acetate.	
(iii)	The extinguisher composed of monoammonium phosphate with a nitrogen carrier.	
(iv)	The extinguisher composed of fine sodium bicarbonate powder pressurized by nitrogen.	
(v)	The extinguisher composed of proteins and fluoro-proteins.	
(vi)	The extinguisher composed of tap water pressurized by air.	

SECTION B: (54 MARKS)
Answer all questions in this section

3. (a) The confusion had occurred at Ngorongoro National Park when the bottle containing substance G was found in a hole where one of the wild animals was living. A chemist ends the confusion by exposing cobalt (II) chloride paper to substance G; the cobalt (II) chloride changes from blue to pink.
- Identify the name of substance G.
 - Outline two chemical properties of substance G.
 - Suggest the other chemical test to identify the presence of substance G, then give its positive observation.
- (b) You are provided with the following fuels: Fire woods, natural gas, charcoal, petrol and kerosene.
- Giving two reasons, explain why fire wood and charcoal which are cheap and most available fuels are not recommended for domestic uses.
 - Why is kerosene often recommended to be used as a liquid fuel in school laboratory rather than petrol?
4. (a) Define the term **"Mole"** as used in chemistry.
- (b) Calculate the percentage composition by mass of Nitrogen in Ammonium Nitrate (NH_4NO_3). [Atomic weights: N = 14, H = 1, O = 16]
- (c) What is the molarity of a solution containing 5.3 g of sodium carbonate (Na_2CO_3) dissolved in 250 cm^3 of solution?
5. A sample of impure silver of mass 3.45g was used as the anode in an electrolysis purifying process. The cathode was made up of pure gold of mass 6.45g. After the electrolysis the cathode was found to weigh 9.66g
- Calculate the number of coulombs of electricity used
 - What is the percentage purity of impure silver?

6. (a) Metals are extracted from their ores. Using two important metals in the table below, identify the substance(s) apart from their chief ores which are added during the extraction process and finally, suggest their functions.

Name of the metal to be extracted	Substance(s) added during extraction	Function(s) of the substance(s)
Sodium		
Iron		

- (b) Consider the reaction between aqueous solutions of silver nitrate and calcium chloride.
 (i) With state symbols, write the balanced molecular equation for this reaction.
 (ii) Give three usefulness of the equation in (b)(i) over word equation.

7. (a) Suppose you went to visit your friend in their school and found students arguing with some facts. Help them by providing the reasons on these facts below:

- Cold foods do not smell from a distance
- Cooking pots, pans and utensils are made up of metals.
- Saturated sugar solution can dissolve more sugar after being heated.
- It is not advised to use water to extinguish class B fire.
- Liquids and gases are categorized as fluids.

- (b) After two months some parts with iron materials on Kamara's house appeared to form a permanent reddish-brown coating. Explain four (4) methods that could be used by Kamara to prevent that substance formed on iron materials in his house.

8. (a) You are provided with elements M and N having atomic numbers 12 and 17 respectively.

- Write the chemical formula of the compound formed by the combination of M and N.
- Mention the type of bond by the combination of elements M and N.
- State three properties of the compound formed by the combination above between elements M and N.

- (b) Below are the names of some organic compounds which were named incorrectly:

- But- 3 - Ene
- 1, 1, 1, 1, - tetramethyl methane
- 2,3-methylybutane
- 3-chloro-2-methyl butane

By Using IUPAC rules study the names then write them correctly.

SECTION C: (30 MARKS)

Answer two questions from this section.

9. What are the **five (05)** immediate efforts to be done by the National Environment Management Council (NEMC) in Tanzania in order to ensure that land is not polluted?
10. The Contact Process is used for the large-scale manufacture of Sulphuric acid (H_2SO_4).
 (a) State the three essential conditions (Temperature, Pressure, and Catalyst) for the production of Sulphur trioxide in this process.
 (b) Why is Sulphur trioxide dissolved in concentrated Sulphuric acid to form **Oleum** instead of dissolving it directly in water?
 (c) List two major uses of Sulphuric acid in the industry.
11. The agricultural officer at Usangu ward conducted a meeting with all farmers in that ward todiscuss the activities which cause the loss in soil fertility in their land. Describe any six (6) possible ways which were mentioned by the officer at that meeting.